

SMART INDIA HACKATHON 2026



Problem Statement ID – SIH26125

Problem Statement Title - Air Power: Dynamic Air Operations & Resource Optimisation System

Theme - Defence & Aerospace / AI

PS Category - Software

Organization - Indian Air Force (IAF)

Team ID - 158358

Team Name - TrustForge-I



AEROOPT AI

DYNAMIC AIR OPERATIONS & RESOURCE OPTIMISATION PLATFORM (IAF DSS)

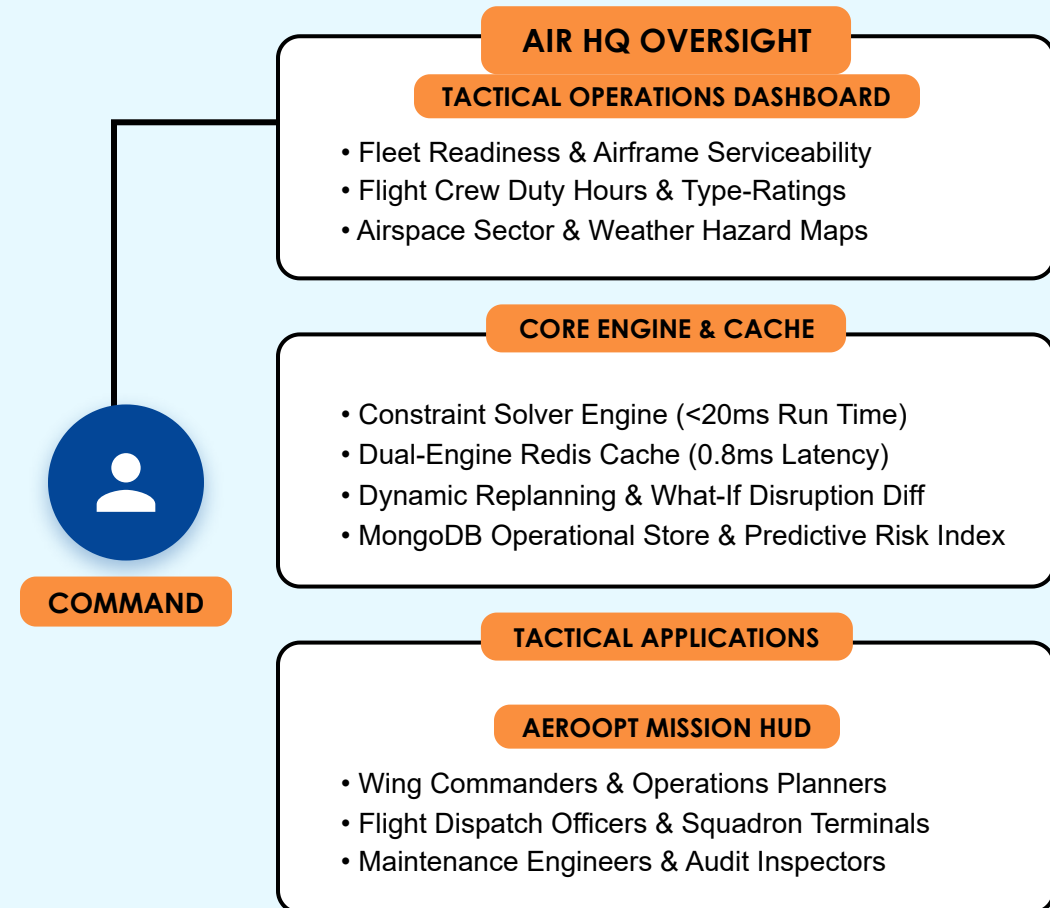
PROBLEM OVERVIEW

- **Fragmented Operational Awareness** – Fleet readiness, aircrew fatigue, weather corridors, and task queues are siloed across disparate legacy dispatch logs.
- **Suboptimal Sortie Scheduling** – Manual dispatch processes cause severe scheduling bottlenecks and fail to recompute allocations during tactical surges.
- **Pilot Fatigue & Regulation Violations** – Lack of automated tracking for cumulative 7-day flight hours risks pilot exhaustion and breaches aviation safety rules.
- **Convective Weather Vulnerabilities** – Static planning cannot dynamically gate weather-sensitive sorties from restricted operational sectors.
- **Vulnerability to Data Tampering** – Flight logs and operational dispatch decisions lack immutable audit trails required for command oversight.

PROBLEM SOLUTION

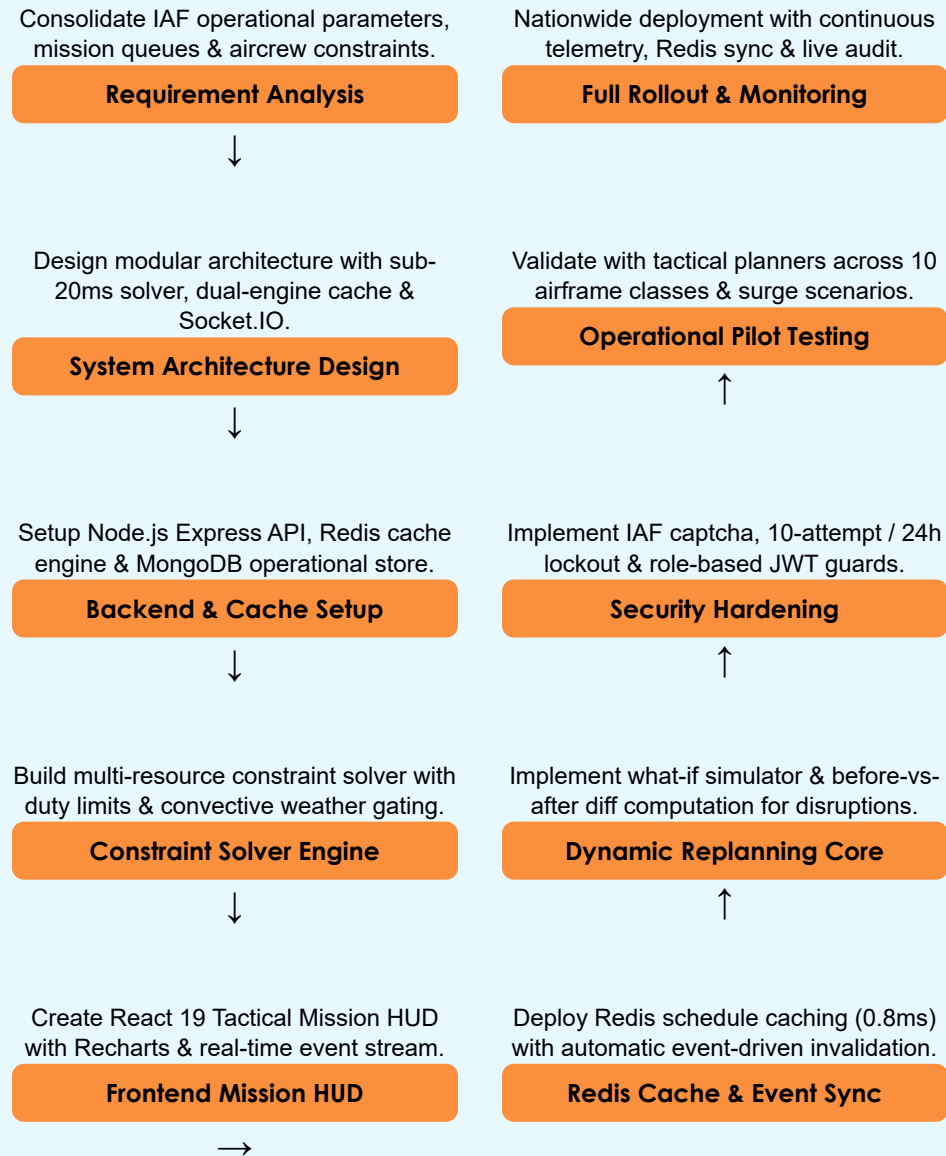
- **Constraint-Based Optimiser** – High-performance solver computes multi-resource feasibility across fleet and aircrew in **< 20 milliseconds**.
- **Redis Dual-Engine Architecture** – Sub-millisecond schedule caching (**0.8ms**) with atomic 24h lockout keys and zero-crash embedded fallback.
- **Dynamic What-If Replanning** – Instant simulation comparing before-vs-after diffs upon airframe grounding, pilot illness, or storm incursions.
- **Automated Weather Corridors** – Live convective weather gating automatically blocks weather-sensitive missions from restricted airspace sectors.
- **Pilot Fatigue Governance** – Strict enforcement of 60-hour weekly duty caps, rest windows, and predictive 72-hour fatigue indices.
- **Human-in-the-Loop Audit Ledger** – Commanders retain absolute operational authority with an immutable audit ledger recording every decision diff.

HIERARCHICAL DIAGRAM

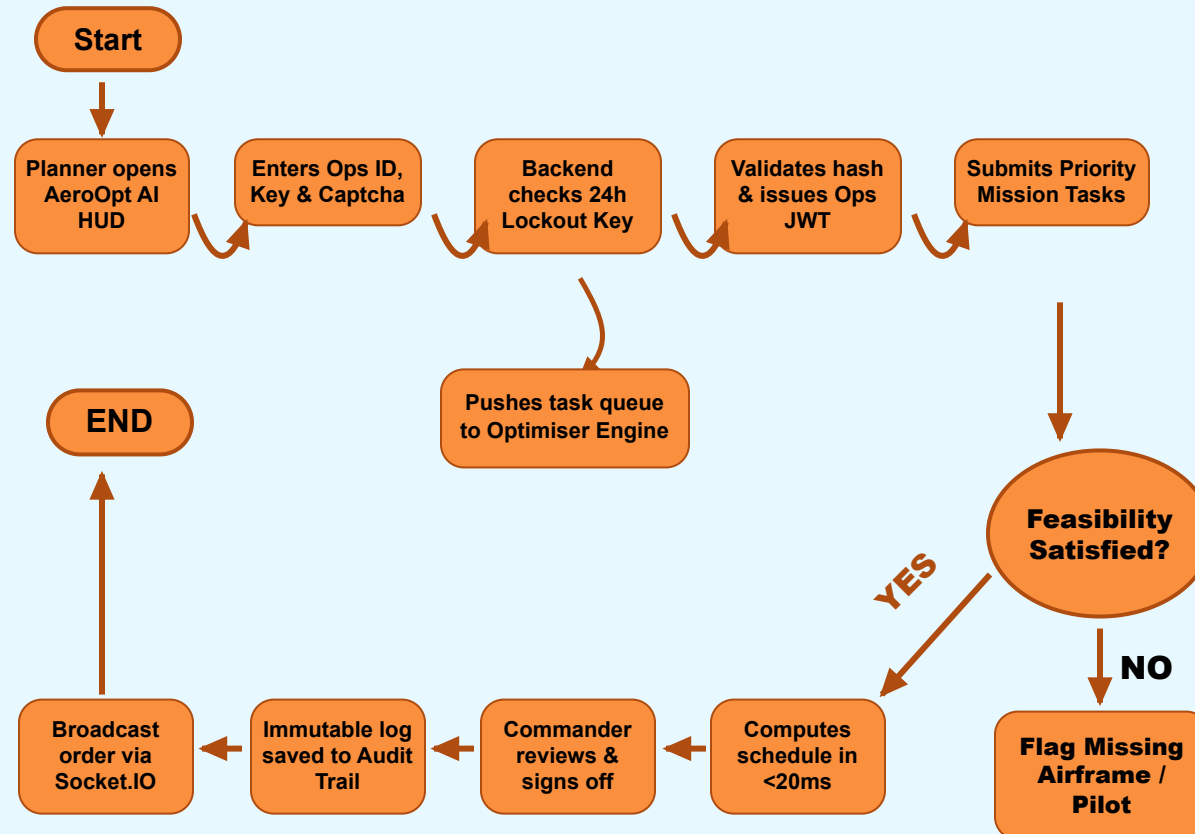


TECHNICAL APPROACH

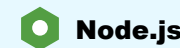
METHODOLOGY



FLOW CHART:



TECH
STACK



FEASIBILITY AND VIABILITY

ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY	2. ECONOMICAL FEASIBILITY	3. SOCIAL FEASIBILITY	4. LEGAL & COMPLIANCE FEASIBILITY	5. OPERATIONAL FEASIBILITY	6. SECURITY FEASIBILITY
Sub-20ms constraint engine, Redis dual-engine caching, and modern React 19 SPA built on proven, production-grade open-source components with zero esoteric dependencies.	Self-hosted architecture eliminates recurring commercial SaaS licensing fees. Dual-engine fallback allows deployment on standard hardware without expensive infrastructure.	Fosters deep trust among flight crews and operations planners by mathematically enforcing pilot rest regulations, eliminating fatigue-induced safety hazards.	Strictly complies with Indian Military Aviation regulations, DGCA flight duty time limitations (FDTL), and CERT-In sovereign cybersecurity directives.	High-contrast IAF tactical HUD designed for high-stress dispatch rooms with zero learning curve, automated what-if simulations, and one-click schedule approvals.	Role-based access control, cryptographic JWT session tokens, 10-attempt/24h terminal lockout via atomic Redis keys, and an immutable audit trail.

POTENTIAL CHALLENGES AND RISKS

HIGH-TEMPO SORTIE BURSTS

Simultaneous surge taskings during conflict escalations risking schedule latency or server bottlenecks.



USER ADOPTION & RESISTANCE

Legacy dispatch operators and flight clerks resisting automated algorithmic recommendations.



SUDDEN CONVECTIVE WEATHER

Dynamic storm fronts rapidly closing operational flight corridors and invalidating schedules.



UNAUTHORIZED DISPATCH ESCALATION

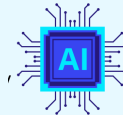
Malicious administrative tampering or credential compromise attempting unauthorized flight orders.



STRATEGIES FOR OVERCOMING THESE CHALLENGES

SUB-20MS CONSTRAINTS & CACHE

High-speed constraint matcher cached in Redis (0.8ms retrieval) handling surge scheduling instantly.



AWARENESS & INTERACTIVE HUD

Zero-learning-curve IAF tactical interface with transparent constraint explanations and guided onboarding.



DYNAMIC REPLANNING SIMULATOR

What-if disruption engine replanning sorties and rerouting flights around severe weather in seconds.



ACCESS CONTROL & AUDIT TRAIL

Strict least-privilege RBAC, 24h lockout policy, and tamper-proof immutable audit logging for all actions.



IMPACT AND BENEFITS

UNIQUE FEATURES:

Sub-20ms Constraint Engine

Solves complex multi-resource flight schedules across 10 airframe classes and crew rosters in milliseconds.

Dynamic What-If Simulator

Immediate before-vs-after diff comparisons for sudden airframe groundings, crew illnesses, or storms.

Redis Dual-Engine Resilience

Sub-millisecond schedule cache with embedded zero-config fallback guaranteeing 100% demo availability.

Automated Weather Gating

Real-time convective storm restriction checks blocking weather-sensitive sorties from high-risk sectors.

Human-in-the-Loop Sign-off

Autonomy with accountability: AI recommends optimal sortie orders while commanders retain approval authority.

IMPACTS

Operational Tempo Acceleration



Cuts flight dispatch scheduling cycle from hours to milliseconds.

Enhanced Mission Readiness



Optimizes fleet serviceability rates through intelligent airframe hour distribution.

Zero-Fatigue Flight Safety



Prevents pilot exhaustion by enforcing strict 60h weekly regulatory duty limits.

Rapid Disruption Recovery



Minimizes mission cancellations during sudden adverse weather or airframe AOG events.

Sovereign Defence Self-Reliance



Completely homegrown decision-support software deployed within Indian borders.

BENEFITS

Tactical Situational Picture



Consolidated single-pane-of-glass HUD displaying fleet, crew, weather, and sorties.

Maximized Fuel & Asset Life



Balances airframe flight hours and fuel reserves to extend depot maintenance intervals.

Audit & Command Accountability



100% transparent decision ledger recording every allocation, approval, and rejection.

Scalable to Tri-Service Ops



Architecture effortlessly scales to army aviation, naval air arms, and joint commands.

Zero Recurring Licensing Costs



Eliminates reliance on expensive foreign military logistics software.

OUR PROTOTYPE

WebApp Interface



Web Interface

RESEARCH AND REFERENCES

- **Operations Research & MILP in Air Operations Scheduling (NATO STO / IEEE)**

Summary: Formulates mixed-integer linear programming (MILP) and constraint satisfaction algorithms for optimal sortie scheduling, airframe turnaround minimization, and crew fatigue mitigation.

Link: <https://www.sto.nato.int/publications/>

- **ICAO Annex 6 & DGCA Flight and Duty Time Limitations (FDTL)**

Summary: International and Indian statutory guidelines governing pilot duty cycles, mandatory rest intervals, and cumulative 7-day fatigue thresholds for flight safety.

Link: <https://www.dgca.gov.in/digigov-portal/>

- **Real-Time In-Memory Caching Strategies for High-Throughput Tactical Systems (ACM)**

Summary: Architectural framework evaluating sub-millisecond in-memory caching, atomic key lockouts, and event-driven invalidation in defense command-and-control grids.

Link: <https://dl.acm.org/doi/10.1145/3318464.3389700>

- **Human-in-the-Loop AI Decision Support for Mission-Critical Command Systems (DARPA)**

Summary: Principles for designing explainable, human-governed AI advisory systems ensuring tactical autonomy operates under strict commander oversight and tamper-proof auditing.

Link: <https://www.darpa.mil/program/explainable-artificial-intelligence>